Core Mathematics

A Practical Guide on Mathematics for Physics Learners

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These are lecture notes for first-year university students in physics or physical science. They cover the mathematical tools you need most: calculus, logic, complex numbers, and vectors, with introductions to vector calculus, differential equations, and statistical analysis. Exercises are central to these notes. Practice each topic repeatedly until you can work through problems quickly and accurately.

Preface

Physics is an activity to describe the world, but how? At present, mathematics (+ *broken* English) is the only language we can use. So, unfortunately, we need to learn mathematics! (and English...🥺) But, in some sense, we are fortunate because we have a language to describe the world. In fact, math and physics have developed together. We understand physics better if we study mathematics well!

When you do DIY, you need to use the tools properly, smoothly, and accurately. It is the same here; when we use math as a tool, it must be **logical**, **quick**, and **accurate**. To be logical, you must think each problem carefully; otherwise your discussion might be incomplete and you can't convince others. To be quick and accurate, you must *"drill"* repeatedly, as if top NBA players do shooting practice almost everyday. You should reach the point where basic calculations feel automatic.

This is not a math textbook. **This is rather** a math "drill" book. Rigorous proofs are mostly omitted, but drill problems are the core of this document, through which students are expected to achieve better conceptual understanding.

Target of this document

This document is primarily for first-year undergraduate students in their second semester, who

- completed basic calculus (differentiation and integration),
- are going to major in physics or physical science, and
- do not hesitate to do repetitive practice to achieve better conceptual understanding.

How to use this document

1. Buy a **A4-sized paper notebook**.

   Digital notebooks on tablets are not suitable (or less effective) for this document.

2. Read the text with solving Quizzes.

   You can solve "Quizzes" without writing down the solution process.

3. Solve Drill Problems to nourish fluency.

   Mind the fluency ($\neq$ speed) and accuracy. Write down the solution process clearly.

4. Solve other problems to have better conceptual understanding (& writing skills).

   Think carefully and write down the solution process clearly as an **English text**, taking care of logical completeness.

5. Optionally, you are encouraged to prepare a reference book, so that you can consult it when you want to know more. (Sho supposes [Boas] listed below is the best option.)

References

This document is written under the influence of the following references.

- The Math Book by Hal Tasaki (数学：物理を学び楽しむために, in Japanese)

  A free book with full description, but unfortunately in Japanese. Topic selections and the depth of descriptions are based on this reference.

- First-Year Math Practice (in Japanese)

  This document is grounded in a "learn steadily" approach ("じっくり" in Japanese), which originates from Prof. Gocho and Prof. Kiyono, who were instructors of Sho in his freshman. Problems and rigorous math descriptions are taken from their works.

- **[Boas]** Mary L. Boas, *Mathematical Methods in the Physical Sciences*, 3rd ed., Wiley, 2006.
- **[AWH]** George B. Arfken, Hans J. Weber, and Frank E. Harris, *Mathematical Methods for Physicists*, 7th ed., Academic Press, 2023.

  These two books are widely used in physics departments around the world. [Boas] seems more friendly to beginners, while [AWH] seems more detailed and complete. Sho recommends you to study [Boas] *in parallel with* this document, using it as a reference, and to consult [AWH] when you want more details or more exercises.

Lecture Plan

The content is designed for 150-minute × 14-week lectures, as it is originally for a lecture *Mathematics for Fundamental Physics* (基礎物理數學) in National Sun Yat-sen University (國立中山大學).

1. Derivative
2. Units. Significant Digits.
3. Logic. Theorems. a^x and $\log_a x$.
4. Logic. Theorems. a^x and $\log_a x$.
5. Vectors are arrows.
6. $a \cdot b$ and $a \times b$
7. Matrices (rotation, reflection, scaling)
8. Complex numbers (polar form, Euler's formula, multi-valuedness)
9. Complex vectors, Hermitian inner product, bra-ket notation
10. Vector calculus (grad, div, rot; example of point charge)
11. ODE basics
12. ODE basics
13. Probability and error analysis
14. Probability and error analysis

♣TODO♣a bit on (abstract) vector space to prep for QM?

Symbols

Mathematical Notations

Chapter 1	
$\mathbb{N}$	natural numbers
$\mathbb{N}_0$	natural numbers (0, 1, 2, ...)
$\mathbb{N}^+$	natural numbers (1, 2, ...)
$\mathbb{R}$	real numbers
$\mathbb{Q}$	rational numbers
$\mathbb{C}$	complex numbers

Greek symbols

A	α	alpha	H	η	eta	N	ν	nu	T	τ	tau
B	β	beta	Θ	θ	theta	Ξ	ξ	xi	Y	υ	upsilon
Γ	γ	gamma	I	ι	iota	O	o	omicron	Φ	φ (ϕ)	phi
Δ	δ	delta	K	κ	kappa	Π	π	pi	X	χ	chi
E	ε	epsilon	Λ	λ	lambda	P	ρ	rho	Ψ	ψ	psi
Z	ζ	zeta	M	μ	mu	Σ	σ	sigma	Ω	ω	omega

Gray-outed ones are almost never used, but others are almost always used. You need to write them so that **others can distinguish each from others**, but Sho thinks Japanese- or Chinese-speakers are OK with it, as we everyday read and write 日 vs 曰 vs 白 or 土 vs 士 vs 工 properly!

Quiz

★**1.** Read out the following words.

(1) α-particle **(2)** β-decay **(3)** Γ-function

(4) ε-δ definition **(5)** angle θ and φ **(6)** μ- and τ-leptons

(7) $\psi(x)$ and $\varphi(x)$ **(8)** ζ and ξ **(9)** metric $\eta_{\mu\nu}$

(10) X-ray and γ-ray **(11)** Ω-baryon **(12)** $\varepsilon_{\mu\nu\rho\sigma}$-tensor

Problems

💪**0.1.** Write the following letters so that people can distinguish each from others.

(1) a, α **(2)** b, β **(3)** c, C **(4)** e, E, ε

(5) ζ, ξ **(6)** $\Theta, \theta, Q, \sigma, 6$ **(7)** $g, q, 9$ **(8)** $i, l, 1, I$

(9) k, K, κ	**(10)** m, μ	**(11)** p, ρ	**(12)** s, S
(13) t, τ, T	**(14)** u, v, ν	**(15)** w, W, ω	**(16)** x, X, χ

SI Units

Historically, different civilizations used different units. Today, science and engineering worldwide use the **SI units** (Système International d'Unités)—the single international standard. It is built from seven **SI base units**:

Table 1: SI base units and their symbols for dimension [1]. Recall that units are case-sensitive.

quantity	symbol and name		symbol for dimension
time	s	(second)	T
length	m	(meter)	L
mass	kg	(kilogram)	M
electric current	A	(ampere)	I
temperature	K	(kelvin)	Θ
amount of substance	mol	(mole)	N
luminous intensity	cd	(candela)	J

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Bibliography

[1] Bureau International des Poids et Mesures, "The International System of Units (SI)," BIPM. [Online]. Available: https://doi.org/10.59161/AUEZ1291

Chapter 1

Derivative (Review)

1.1 The first step

Remark: This document is designed for first-year students in their *second semester*, who **have already studied derivatives** in their first-semester *“calculus”* course. If you are not yet confident with derivatives, please study the Derivative Boot Camp first[#1].

Birds sing, fish swim, flowers bloom, stars twinkle, and university students calculate derivatives. Let us begin with basic **derivatives**. Try the next quiz—and note how long it takes.

Quiz

★**1.** Calculate the first derivatives of the following functions.

(1) $(x+1)^3$ **(2)** $\tan x$ **(3)** $2\cos^2 x$ **(4)** $2\cos x^2$

(5) $x^3 \cos x$ **(6)** $\dfrac{3\sin x}{x}$ **(7)** $x^{3/2}$ **(8)** $\sqrt{\sin x}$

If you finished within two minutes—Wonderful! Most students take 3–4 minutes. If you took more than four minutes, do not worry: you can do the calculations, and a little more practice will help. Try some problems in the Derivative Boot Camp

Did you make calculation mistakes and get wrong answers? That is a serious issue—just as serious as forgetting how to do the calculations at all. You will perform similar calculations more than 100 times in university, so you need to be both fast and accurate.

At university, students often underestimate the importance of basic calculations. In physics, simple calculations appear constantly, so your speed and accuracy directly affect how well you follow lectures, how efficiently you study, and ultimately your grade. Both can be improved with practice.

This course will help you strengthen these basic calculations—some of which you already know from high school—while also introducing new topics that are essential for physics. You will solve many problems and drills, just as a beginner athlete repeats the same move hundreds of times to master it.

[#1] Visit `https://misho104.github.io/LecturePublic` and find `gp1_boot1_deriv_true.pdf`.

1.2 Typical Pitfalls

1.2.1 Notations

There are several equivalent ways to write the derivative of $f(x)$:

$$f'(x) \quad = \quad \frac{\mathrm{d}f}{\mathrm{d}x}(x) \quad = \quad \frac{\mathrm{d}f(x)}{\mathrm{d}x} \quad = \quad \frac{\mathrm{d}}{\mathrm{d}x}f(x) \qquad \text{[All means the same thing].} \tag{1.1}$$

Similarly, the value of $f'(x)$ at a specific point $x = 3$ can be written as

$$f'(3) \quad = \quad f'(x)\Big|_{x=3} \quad = \quad \frac{\mathrm{d}f}{\mathrm{d}x}\bigg|_{x=3} \quad = \quad \frac{\mathrm{d}f}{\mathrm{d}x}(3) \quad = \quad \frac{\mathrm{d}}{\mathrm{d}x}f(3), \tag{1.2}$$

and they are all equivalent. For example, if $f(x) = 3x^2 + 6$, then $f'(x) = 6x$ and $f'(3) = 18$.

The real pitfalls come next. Students often get confused when a **constant** a appears. If a is *declared as a constant*, we can define a function such as $g(x) = ax^2 + 2a$, for which

$$g'(x) = 2ax. \tag{1.3}$$

We can evaluate $g'(x)$ at $x = a$. The result is $g'(a) = 2a^2$, written as

$$g'(a) \quad = \quad g'(x)\Big|_{x=a} \quad = \quad \frac{\mathrm{d}g}{\mathrm{d}x}(a) \quad = \quad \frac{\mathrm{d}}{\mathrm{d}x}g(a) \quad = \quad 2a^2. \tag{1.4}$$

Be careful: the notation $\frac{\mathrm{d}}{\mathrm{d}x}g(a)$ does **not** mean $\frac{\mathrm{d}}{\mathrm{d}x}\big[g(a)\big] = \frac{\mathrm{d}}{\mathrm{d}x}\big(a^3 + 2a\big) = 0$.

Fail safe note: If you can't see Eq. (1.3), try setting $a = 3$. Then, $g(x) = f(x)$, and $g'(x)$ should equal $f'(x)$.

There is more bad news. *Physicists are often lazy*, and we frequently omit the argument (x). For example, if $f(x) = 3x^2$ and $F(t) = t^2 + at + 1$, and if a is *declared as a constant*,

$$\begin{aligned} f' &= \frac{\mathrm{d}}{\mathrm{d}x}f = 6x, & f'(a) &= \frac{\mathrm{d}}{\mathrm{d}x}f(a) = 6a, \\ F' &= \frac{\mathrm{d}}{\mathrm{d}t}F = 2t + a, & F'(a) &= \frac{\mathrm{d}}{\mathrm{d}t}F(a) = 3a. \end{aligned} \tag{1.5}$$

So when you see a function written as f, you must identify its variable by reading the context. For example, we sometimes replace the variables. Assume $f(x) = 3x^2$. If t is another **variable** (= not a constant), we can understand $3t^2$ is the same function $f(t)$. So we may write $f(t) = 3t^2$, whose derivative is $f'(t) = \mathrm{d}f/\mathrm{d}t = 6t$.

Advanced note: We usually write "Let k be a constant" to declare k as a constant. You need to do so, too.

Higher-order derivatives is written as

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\mathrm{d}f}{\mathrm{d}x}\right) = \frac{\mathrm{d}^2 f}{\mathrm{d}x^2} = f''(x) = f^{(2)}(x), \qquad \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\mathrm{d}f}{\mathrm{d}x}\right)\right) = \frac{\mathrm{d}^3 f}{\mathrm{d}x^3} = f'''(x) = f^{(3)}(x), \tag{1.6}$$

etc. Please be careful on the position of "2" and "3".

Quiz

★**2.** Let a and b be constants. If $f(x) = 3x^2$, $g(t) = at^2 + b$, $F(x) = a + 3$, and $G(t) = a\cos(t) + x\sin(t)$, what are the following expressions?

(1) $f'(x)$ **(2)** $f'(0)$ **(3)** $f'(a)$ **(4)** $f'(-a)$

(5) $g'(t)$ **(6)** $g'(0)$ **(7)** $g'(b)$ **(8)** $g'(x)$

(9) $F'(a)$ **(10)** $G'(t)$ **(11)** $G'(0)$ **(12)** $G'(a)$

(13) $\dfrac{\mathrm{d}f}{\mathrm{d}x}$ **(14)** $\dfrac{\mathrm{d}f}{\mathrm{d}x}(0)$ **(15)** $\dfrac{\mathrm{d}g}{\mathrm{d}t}$ **(16)** $\dfrac{\mathrm{d}g(x)}{\mathrm{d}x}$

1.2.2 Radian and trigonometric functions

At university, angles are almost always measured in **radians**:

$$360 \text{ degree} \quad (\text{or: } 360°) \quad = \quad 2\pi \text{ radian} \quad (\text{or: } 2\pi \text{ rad}) \tag{1.7}$$

Furthermore, we usually omit "radian" (because we are lazy!) Now,

$$\text{a right angle is } \pi/2. \quad \text{The sum of the interior angles of a triangle is } \pi. \tag{1.8}$$

Quiz

★**3.** Express the following angles in radians, and radians in angles.

(1) 180° **(2)** 45° **(3)** 60° **(4)** 90 **(5)** −30°

(6) 1° **(7)** 1 **(8)** −0.3 **(9)** $x°$ **(10)** x

Why do insist on radians? The answer comes from the derivative formula

$$\frac{\mathrm{d}}{\mathrm{d}x}\sin x = \cos x. \tag{1.9}$$

Because we have chosen radians as the standard, $(\sin x)'$ becomes this simple.

> **Advanced note:** The simpleness of Eq. (1.9) originates in the fact $\sin(0.01 \text{ rad}) \approx 0.01$. If we used degrees, we would have $\sin(0.01°) \approx 0.01 \times 0.017453$ and everything is messed up with this number 0.017453.

There are a few remarks in the notation of **trigonometric functions**:

$$\begin{aligned} &\sin^2 x \neq \sin x^2. && \text{Namely,} \quad (\sin x)^2 = \sin^2 x && \neq \quad \sin x^2 = \sin(x^2). \\ &\tan^{-1} x \neq \frac{1}{\tan x}. && \text{Namely,} \quad \tan^{-1} x = \arctan x && \neq \quad (\tan x)^{-1} = \frac{1}{\tan x} = \cot x. \end{aligned} \tag{1.10}$$

The following are not incorrect but confusing. *We should avoid ambiguity*, so please don't use them.

$$\sin^{-2} x, \quad \sin^{1/2} x, \quad \sin(x)^2, \quad \sin(x+1)^2, \ldots$$

Sho thinks we should only use $\sin^k x$ for $k = 2, 3, 4, \ldots$, and use $\arcsin x$ instead of $\sin^{-1} x$.

1.2.3 Several interpretations of derivatives

When you, physics learners, discuss $f'(t)$, you should have the following three interpretations:

- Regarding t as the time, $f'(t)$ is the **rate of change** of $f(t)$ per unit time. If $f'(t) > 0$, the function f is increasing at the time t, while $f'(t) < 0$ means it is decreasing.
- Consider a graph of $f(t)$. Then, $f'(t)$ is the slope of the **tangent line** to the graph at the point t.
- Mathematically, $f'(t)$ is defined by (Check that they are equivalent.)

$$f'(t) \coloneqq \lim_{\Delta t \to 0} \frac{f(t+\Delta t) - f(t)}{\Delta t} = \lim_{h \to 0} \frac{f(t+h) - f(t-h)}{2h} = \lim_{s \to t} \frac{f(s) - f(t)}{s - t}. \tag{1.11}$$

If you are unsure of them, please consult your first-year Calculus textbooks for further information.

1.3 Taylor expansion

Reviewing the previous equation for a function $f(x)$, we have

$$f'(x) \coloneqq \lim_{\Delta x \to 0} \frac{f(x+\Delta x) - f(x)}{\Delta x}, \qquad f'(a) \coloneqq \lim_{\delta \to 0} \frac{f(a+\delta) - f(a)}{\delta}. \tag{1.12}$$

> **Advanced note:** These two have a bit different sense: the first one defines a new function $f'(x)$, while the second one defines a number that is eventually equal to $f'(x)|_{x=a}$. Anyway, we don't care the difference.

We can understand the second equation as

$$\text{If } \delta \approx 0, \quad f'(a) \approx \frac{f(a+\delta) - f(a)}{\delta}, \quad \text{i.e.,} \quad f(a+\delta) \approx f(a) + \delta f'(a). \tag{1.13}$$

These equations will be helpful if you want to know **what is $f(x)$ around the point $x = a$?**

Example 1.1

Find the approximate value of the following expressions:

(1) $\sqrt{1.002}$ **(2)** $(1.002)^{10}$ **(3)** $\sqrt{4.004}$

Solution

(1) Apply Eq. (1.13) for $f(x) = \sqrt{x}$, $a = 1$, and $\delta = 0.002$. Then,

$$f'(x) = \frac{1}{2\sqrt{x}}, \quad f'(a) = f'(1) = \frac{1}{2}, \quad f'(a) \approx f(a) + \delta f'(a) = \sqrt{1} + \frac{\delta}{2} = \underline{1.001}.$$

(2) Doing the same thing for $g(x) = x^{10}$, we have

$$g'(x) = 10x^9, \quad g'(a) = g'(1) = 10, \quad g'(a) \approx g(a) + \delta g'(a) = 1^{10} + 10\delta = \underline{1.02}.$$

(3) Doing the same thing for $h(x) = \sqrt{x}$, $a = 4$, and $\delta = 0.004$, we have

$$h'(4) = \frac{1}{4}, \quad h'(4.004) \approx h(4) + 0.004 \times \frac{1}{4} = \underline{2.001}.$$

This technique, and Eq. (1.13), is a special case of the more general **Taylor's theorem**:

♣TODO♣postpone the discussion below to a later chapter

Theorem 1.2 (Taylor's theorem)

If a function $f(x)$ is sufficiently smooth, we can express it as

$$f(x) = \sum_{k=0}^{N} \frac{(x-a)^k}{k!} f^{(k)}(a) + [\text{dangerous part}]. \tag{1.14}$$

and we can ignore the dangerous part if $x \approx a$. So, if we replace x by $a + \varepsilon$ and $\varepsilon \approx 0$,

$$\begin{aligned} f(a+\varepsilon) &= \sum_{k=0}^{N} \frac{\varepsilon^k}{k!} f^{(k)}(a) + [\text{dangerous part}] \\ &\approx f(a) + \varepsilon f'(a) + \frac{\varepsilon^2}{2} f''(a) + \frac{\varepsilon^3}{6} f'''(a) + \cdots \end{aligned} \tag{1.15}$$

Problems

★**1.1.** Find the approximate value of the following expressions:

(1) $(1.002)^5$ **(2)** $\sin(0.001)$ **(3)** $\cos(0.001)$

(4) $\tan(0.001)$ **(5)** $\sqrt{4.004}$

*****1.2.** Find the approximate value of the following expressions:

(1) $\sin(3.14)$ **(2)** $\cos(1.57)$ **(3)** $\sqrt[3]{1.001}$

(4) $(1.002)^{0.9}$ **(5)** $1/(10.004)^3$ **(6)** $\sqrt{10001}$

Fail safe note: If you are not confident with the meaning of $\sqrt[3]{1.001}$, $(1.001)^{-0.9}$, etc., please go to ♣TODO♣a chapter...

****1.3.** Write down the definition of the second derivative $f''(x)$. Repeat the discussion of Eq. (1.13) to find the expansion $f'(a+\varepsilon) \approx f'(a) + \varepsilon f'(a) + (\varepsilon^2/2) f''(a)$.

♣TODO♣more problems

Chapter 2

Units and Significant figures

A **physical quantity** is usually made of a number, a unit, and an **uncertainty** (**error**). For example,

	unit	central value	absolute uncertainty	relative uncertainty
$10\,\mathrm{m} \pm 1\,\mathrm{cm}$	m (meter)	$10\,\mathrm{m}$	$1\,\mathrm{cm} = 0.01\,\mathrm{m}$	0.001 (or 0.1%)
$1.6\,\mathrm{A} \pm 0.04\,\mathrm{A}$	A (ampere)	$1.6\,\mathrm{A}$	$0.04\,\mathrm{A}$	0.025 (or 2.5%)
$(5 \pm 1) \times 10^{-3}\,\mathrm{C}$	C (coulomb)	$0.005\,\mathrm{C}$	$0.001\,\mathrm{C}$	0.2 (or 20%)
$(50 \pm 1)\,\mathrm{m/s}$	m/s	$50\,\mathrm{m/s}$	$1\,\mathrm{m/s}$	0.02 (or 2%)

Measurements are always with uncertainty. Proper Handling of the uncertainty is a fundamental skill. Uncertainty can often be more important than the central value.

> **Remark:** Upper- and lowercase letters are distinguished. “M” **does not** mean “meter”. The unit “A” is “ampere”, not “Ampere”. The unit “coulomb” is “C”, not “c”.

Quiz

★**1.** For each of the following values, find its unit, central value, absolute uncertainty, and relative uncertainty.

(1) $50\,\mathrm{kg} \pm 500\,\mathrm{g}$ **(2)** $(0.05 \pm 0.001)\,\mathrm{A}$

(3) $(1.6 \pm 0.001) \times 10^{-19}\,\mathrm{C}$ **(4)** $72\,\mathrm{km/h} \pm 1\,\mathrm{m/s}$

★**2.** The next passage has six (6) errors in the use of uppercase and lowercase letters. Find them out.

> In the SI system, temperature is expressed in k (kelvin), a unit named after Lord Kelvin. The unit of force is N (newton), named after the british scientist Isaac newton. The units A (Ampere) and c (coulomb) are named after French scientists. In contrast, Kg (kilogram) is not named after a person, but comes from Greek.

As an undergraduate student, you need to follow the following rules:

1. Numbers are always with units, even in calculations.
2. Include units in a symbol.
3. For physical quantities, use decimals (e.g., 1.6 A). Do not use fractions like (8/5) A.
4. Use significant figures to express the measurement precision.

Significant figures are a simple way to express the uncertainty. We discuss it in this chapter, while Chapter ♣TODO♣chapter has further discussion on uncertainty and error analysis.

2.1 Units

When we educate kids, we write "my height is $\boxed{}$ cm", or "my height is h cm", where $h = 172$ is just a number. But this is not nice! We want to convert the units freely and write equations such as $1.72\,\mathrm{m} = 172\,\mathrm{cm} = 0.00172\,\mathrm{km}$. So,

we always include units in symbols: $\boxed{h = 172\,\mathrm{cm}}$

and then

we can write: $\boxed{h = 172\,\mathrm{cm} = 1.72\,\mathrm{m} = 0.00172\,\mathrm{km} = 1.82\times10^{-16}\,\text{light-year.}}$

Similarly, if $m = 110\,\mathrm{g}$ and $\mathrm{g} = 9.8\,\mathrm{m/s^2}$,

we may write: $w = mg = 110\,\mathrm{g} \times 9.8\,\mathrm{m/s^2} = 1.1\,\mathrm{kg{\cdot}m/s^2}$

but not: $w = mg = 0.11 \times 9.8 = 1.1\,\mathrm{kg{\cdot}m/s^2}$

This second equation is incorrect because m is not equal to 0.11; m is equal to 0.11 kg or 110 g.

Quiz

💪**3.** **(1)** If $mr\omega^2 = 10.0\,\mathrm{kg{\cdot}m/s^2}$, $m = 5.0\,\mathrm{kg}$, and $r = 1.0\,\mathrm{m}$, what is ω?

(2) If $v_0 t + \frac{1}{2}at^2 = 5.0\,\mathrm{m}$, where $a = -4.0\,\mathrm{m/s^2}$ and $v_0 = 7.0\,\mathrm{m/s}$, what is t?

(3) At time $t = 0$, a particle is at $(x, y) = (2.0\,\mathrm{m}, 0)$. It moves with a constant velocity $(v_x, v_y) = (3.0, 4.0)\,\mathrm{m/s}$. What is its position at $t = 1.0\,\mathrm{s}$?

Remark: Usually, physicists use upright fonts for units and *italic fonts* for quantities. For example, m, T, and C are quantity symbols (describing mass, temperature, etc.), while m, T, and C are units (meter, tesla, and coulomb, respectively).

Every physical concept has its own unit. For example, speed has m/s, acceleration has $\mathrm{m/s^2}$, and energy has $\mathrm{kg{\cdot}m/s^2} = \mathrm{N{\cdot}m} = \mathrm{J}$. The table below lists the quantities you have learned. Notice that angle (rad) has no dimension. It is called a **dimensionless** quantity.

concept	dimension	typical unit	
speed	$\mathsf{L\,T^{-1}}$	m/s	
acceleration	$\mathsf{L\,T^{-2}}$	$\mathrm{m/s^2}$	
linear momentum	$\mathsf{M\,L\,T^{-1}}$	kg·m/s	
force	$\mathsf{M\,L\,T^{-2}}$	$\mathrm{kg{\cdot}m/s^2}$	(= N)
energy / work	$\mathsf{M\,L^2\,T^{-2}}$	$\mathrm{kg{\cdot}m^2/s^2}$	(= J)
frequency	$\mathsf{T^{-1}}$	1/s	(= Hz)
angle	1 (dimensionless)	rad	

Advanced note: To understand why angle has no dimension, you may consider the definition of the radian: it is defined as the ratio of the arc length to the radius, so the units cancel (meter / meter = 1).

Here,

please do not write: the force is kg·m^2/s^2, which is also called newton (N).

because *we cannot mix units and concepts*. A correct (and easy) way is to use English words:

the unit of the force is kg·m/s^2 or the force has the unit kg·m/s^2.

Similarly, the following statement is incorrect because it mixes concepts and units:

Since $m = \mathrm{kg}$ and $g = \mathrm{m/s^2}$, $mg = \mathrm{kg{\cdot}m/s^2} = \mathrm{N}$.

Instead, you need to write

Since m has a unit of kg and g has a unit of m/s^2, mg has a unit of $\mathrm{kg{\cdot}m/s^2} = \mathrm{N}$.

Notice the last equation: units can be equal to other units, so we may write

$$\mathrm{N} = \mathrm{J/m} = \mathrm{kg{\cdot}m/s^2}, \qquad \mathrm{m} = \mathrm{J/N}, \qquad \mathrm{s} = \sqrt{\mathrm{kg{\cdot}m/N}}.$$

These are scientifically correct equations.

Remark: We sometimes use informal notations, such as $[F] = \mathrm{N}$ or $F \dashrightarrow \mathrm{N}$ to express "the unit of F is N (newton)". Still, using an equal symbol (=) is **not allowed**. Namely, $F = \mathrm{N}$ is always incorrect.

Quiz

★**4.** Using English, explain what unit the following quantities have.

(1) speed **(2)** velocity **(3)** force **(4)** energy **(5)** power

Advanced note: In very formal situations, we use **symbols for dimensions** (see Table 1 on page 5):

Since $\dim(m) = \mathsf{M}$ and $\dim(g) = \mathsf{L\,T^{-2}}$, $\dim(mg) = \mathsf{M\,L\,T^{-2}}$,

with the operator "dim" [1]. However, it seems too complicated for most situations.

Problems

***__2.1.__ Express the following units only with SI base units, e.g., $\mathrm{N} = \mathrm{kg{\cdot}m/s^2}$.

(1) W (watt) **(2)** Pa (pascal) **(3)** J (joule) **(4)** C (coulomb)

**__2.2.__ Use "dim" notation to express the dimensions of the following quantities. For example, if v is speed, then $\dim(v) = \mathsf{L\,T^{-1}}$.

(1) speed v **(2)** force F

(3) angular momentum L **(4)** electric charge Q

(5) resistance R

(6) specific heat capacity c

(7) Planck constant h

(8) fine-structure constant α

****2.3.** Find a few more examples of dimensionless quantities in physics.

2.2 Significant figures

In researches, we need to treat uncertainties in the method given in ♣TODO♣chap. However, most of lectures use **significant figures** to express the accuracy of a value, so that students becomes familiar with the concept of uncertainties. For example,

3.14	means the last digit "4" is uncertain. It can be 3.1***6***, 3.1***5***, 3.1***1***, 3.1***0***, ...
3.141	means "3.14" is for sure but it can be 3.14***3***, 3.14***2***, 3.14***0***, 3.1***39***, ...
3.1415	might be 3.141***6***, 3.1415***3***, etc., but the author is sure it **cannot** be 3.145 or 3.135.

You have seen trailing zeros in your textbook. They are important because

1.000***0***	means the last digit "0" is uncertain so it may be 1.0002, 1.0001, 0.9998, ...
1.0***0***	can be 1.03 or 0.99, but the author is sure it **cannot** be 1.2 or 0.8.

This notation is often combined with **scientific notation**:

0.00012***3***	is OK but we prefer	$1.2\boldsymbol{3}\times 10^{-4}$.	Both mean the value can be $1.2\boldsymbol{2}\times 10^{-4}$ etc.
0.0010***0***	is OK but we prefer	$1.0\boldsymbol{0}\times 10^{-3}$.	
2997***9***	is OK but we prefer	$2.997\boldsymbol{9}\times 10^{4}$.	

We need to *avoid ambiguous notations*. Namely,

2040	is ambiguous and not nice. We are not sure the author means 2.04×10^3 or 2.040×10^3.
120	is not nice. We are not sure the author means It may mean 1.20×10^2 or 1.2×10^2.
42000	is not nice because it has four ways to interpret the author's intention.

Quiz

★**5.** What are the four interpretations of 42000? Write them in scientific notation.

★**6.** **(1)** Some of the following numbers are ambiguous. Point them out.
(2) Write the other (not ambiguous) numbers in scientific notation.

152 340 1000 9999 43210 0.0300 0.00213 31.0 31.00 31

★**7.** The expressions 0.123, 1.23, and 123 are numbers with ***three*** *significant figures*. Similarly, 1234 and 1.234×10^{-3} are numbers with ***four*** *significant figures*. How about the following expressions?

(1) 3.14	**(2)** 0.11	**(3)** 1.1×10^{-2}	**(4)** 1.1×10
(5) 1.0008	**(6)** 1.0020	**(7)** 1.0000	**(8)** 4.00×10^{-10}
(9) 0.0008	**(10)** 0.00080	**(11)** 12345	**(12)** 67890

Remark: On computers, we often use `2.99e8` (or `2.99E8`) to mean 2.99×10^8, or `1.610e-12` (or `1.610E-12`) to mean 1.610×10^{-12}. However, we **should not** use them in hand-writing.

We use **rounding**-to-the-nearest (四捨五入) when necessary. For example, if you need to convert to three significant figures, it will be

$$1.234 \to 1.23, \quad 1.235 \to 1.24, \quad 31.98 \to 32.0, \quad 1234 \to 1.23 \times 10^3, \quad \text{and so on.}$$

Quiz

★**8.** Round the following numbers to three significant figures.

(1) 41.11 **(2)** 98.76 **(3)** 100.12 **(4)** 15.449

(5) 2.2360 **(6)** 0.0123456 **(7)** 9876 **(8)** 9.999×10^4

2.3 Calculation with Significant figures

Now we are to calculate these numbers with significant figures, such as $1.23 + 4.56$ or $1.23/4.56$, but how? The professional methods are given in ♣TODO♣chap. Here we discuss a simple method for

- addition $(x + y)$ and subtraction $(x - y)$
- multiplication $(x \times y)$ and division $(x \div y$ or $x/y)$

Treatments for $\sqrt{x}$, e^x or $\sin(x)$ need the professional method given in ♣TODO♣chap.

Multiplication and Division

If x has m significant figures and y has n significant figures, then $x \times m$ or $x \div y$ should be rounded to have $\min(m, n)$ significant figures.

Example 2.3

Calculate the following expressions, using calculators.

(1) 8.912×2.4 **(2)** $5.0 \div 3.14$ **(3)** 2.01×49.8 **(4)** 1.210^3

Solution

Here we use a shorthand notation "3-SF" to mean "three significant figures".

(1) Since 8.912 has 4-SF and 2.4 has 2-SF, we round the result to (the smaller) 2-SF:

$$8.912 \times 2.4 = 21.3888 \to \underline{21}.$$

(2) Since 5.0 has 2-SF and 3.14 has 3-SF, we round the result to 2-SF:

$$5.0 \div 3.14 = 1.592\ldots \to \underline{1.6}.$$

(3) Both 2.01 and 49.8 have 3-SF, so we keep 3-SF, but don't forget to avoid ambiguity!

$$2.01 \times 49.8 = 100.098 \to 100 \to \underline{1.00 \times 10^2}.$$

(4) $1.210^3 = 1.210 \times 1.210 \times 1.210$, so we round the result to 4-SF:

$$1.210^3 = 1.771561 \rightarrow \underline{1.772}.$$

This treatment is justified if you carry out the long multiplication by your hand.

$$\begin{array}{cccccc} & & & 1. & 2 & \mathit{0} \\ \times & & & 1. & 3 & \mathit{1} \\ \hline & & & \mathit{1} & \mathit{2} & \mathit{0} \\ & & 3 & 6 & \mathit{0} & \\ & 1 & 2 & \mathit{0} & & \\ \hline & 1 & 5 & \mathit{7} & \mathit{2} & \mathit{0} \end{array} \qquad \begin{array}{ccccccc} & & & 8. & 9 & 1 & \mathit{2} \\ \times & & & & & 2. & \mathit{4} \\ \hline & & \mathit{3} & \mathit{5} & \mathit{6} & \mathit{4} & \mathit{8} \\ & 1 & 7 & 8 & 2 & \mathit{4} & \\ \hline & 2 & \mathit{1} & \mathit{3} & \mathit{8} & \mathit{8} & \mathit{8} \end{array} \qquad \begin{array}{ccccccc} & & & 2. & 0 & \mathit{1} \\ \times & & & 4 & 9. & \mathit{8} \\ \hline & & \mathit{1} & \mathit{6} & \mathit{0} & \mathit{8} \\ & 1 & 8 & 0 & \mathit{9} & \\ & 8 & 0 & \mathit{4} & & \\ \hline 1 & 0 & \mathit{0.} & \mathit{0} & \mathit{9} & \mathit{8} \end{array}$$

Observe which digits are uncertain, and how they "pollute" the results in each step. In $1.2\mathit{0} \times 1.3\mathit{1}$, the last $\mathit{1}$ pollutes the part of $\mathit{1} + 6$ and we get an uncertain number $\mathit{7}$. So, the result is $1.5\mathit{7}$.

Addition and subtraction

Different rules are applied for addition and subtraction, where we do long addition and observe which digits are polluted.

Example 2.4

Calculate the following expressions.

(1) $12.3 + 4.56$ **(2)** $123 + 4.56$ **(3)** $0.50 - 0.032$ **(4)** $1.20 \times 10^3 - 27$

Solution

$$\begin{array}{ccccc} & 1 & 2. & \mathit{3} & \\ + & & 4. & 5 & \mathit{6} \\ \hline & 1 & 6. & \mathit{8} & \mathit{6} \end{array} \qquad \begin{array}{cccccc} & 1 & 2 & \mathit{3} & & \\ + & & & 4. & 5 & \mathit{6} \\ \hline & 1 & 2 & \mathit{7.} & \mathit{5} & \mathit{6} \end{array} \qquad \begin{array}{cccccc} & 0. & 5 & \mathit{0} & & \\ - & 0. & 0 & 3 & \mathit{2} \\ \hline & 0. & 4 & \mathit{6} & \mathit{8} \end{array} \qquad \begin{array}{ccccc} & 1 & 2 & \mathit{0} & \mathit{0} \\ - & & & 2 & \mathit{7} \\ \hline & 1 & 1 & \mathit{7} & \mathit{3} \end{array}$$

and we round the results. So, the answers are $16.\mathit{9}$, $12\mathit{8}$, $0.4\mathit{7}$, and $1.1\mathit{7} \times 10^3$.

> **Remark:** These are the beginner rules. Professional scientists and engineers use more nuanced conventions that depend on their field. You will learn those from your supervisors in the future.

Problems

💪**2.4.** Calculate the following, taking care of significant figures. You may use calculators.

(1) 1.23×4.5 **(2)** 1.50×2.0 **(3)** 2.00×5.00 **(4)** 4.0×2.5

(5) $1.23 + 4.5$ **(6)** $1.50 + 2.0$ **(7)** $2.00 + 8.00$ **(8)** $100.0 + 0.123$

(9) $1.23 \div 4.5$ **(10)** $6.0 \div 3.00$ **(11)** $9.00 \div 4.5$ **(12)** $1.50 \div 3.0$

(13) $1.23 - 4.5$ **(14)** $3.10 - 0.10$ **(15)** $3.10 - 0.1$ **(16)** $3.10 - 3.0$

(17) $131 + 69$ **(18)** $131 + 6.9$ **(19)** $131 + 0.69$ **(20)** $1.3 \times 10^2 + 69$

(21) $(1.2 \times 10^5) \times 9.4$ **(22)** $1.2 \div (5.2 \times 10^5)$

(23) $(3.33 \times 10^5) \times (6.3 \times 10^4)$ **(24)** $(3.33 \times 10^5) \times (6.3 \times 10^{-4})$

(25) $(3.33\times10^5)\div(6.3\times10^4)$ **(26)** $(3.33\times10^5)\div(6.3\times10^{-4})$

(27) $1.00\times10^4+2.5\times10^3$ **(28)** $3.00\times10^3-4.50\times10^2$

(29) $2.99\times10^2+0.814$ **(30)** $1.23\times10^{-4}+4.5\times10^{-6}$

***2.5. Calculate the following, taking care of significant figures. You may use calculators.

(1) $1.50\times2.000\times3.50$ **(2)** $8.00\div4.00\div0.25$ **(3)** 5.0^3 **(4)** 2.10^3

(5) $1.0+2.0+3.0\times4.0$ **(6)** $17-8.0-3.0$ **(7)** $3.0^3+1.2$ **(8)** 5.0^3-5

(9) $(1.2+3.45)\times2.1$ **(10)** $(8.0-1.25)\div2.0$ **(11)** $1.2\times3.4+5.6$

(12) $18.0-2.5\times1.2$ **(13)** $1.20\times3.40+5.60$ **(14)** $(5.0\times10^2+3.4)\times1.2$

💪**2.6.** Calculate the following with calculators, taking care of significant figures and units.

(1) $3.1\,\mathrm{m/s}\times1.0\,\mathrm{hour}$ **(2)** $3.1\,\mathrm{m}\div25\,\mathrm{cm}$

(3) $1.5\,\mathrm{kg}\times9.8\,\mathrm{m/s^2}$ **(4)** $1.50\,\mathrm{km}+195\,\mathrm{m}$

(5) $(5.2\,\mathrm{kg/m^3})\times(4.0\,\mu\mathrm{m})^3$ **(6)** $(5.2\,\mathrm{kg/m^3})\times(4.0\,\mu\mathrm{m}^3)$

(7) $1.2\,\mu\mathrm{m}\div2.00\,\mathrm{nm}$ **(8)** $(1.7\times10^{-27}\,\mathrm{kg})\times(3.00\times10^8\,\mathrm{m/s})^2$

(9) $2.000\,\mathrm{kg}\times(3.0\,\mathrm{m/s^2})$ **(10)** $2.000\,\mathrm{kg}\times(3.0\,\mathrm{m/s})^2$

(11) $(1.60\times10^{-19}\,\mathrm{C})\div2.0\,\mu\mathrm{s}$ **(12)** $(1.60\times10^{-19}\,\mathrm{C})\times5.0\,\mathrm{kV}$

(13) $(6.63\times10^{-34}\,\mathrm{J\cdot s})\times(3.0\times10^{12}\,\mathrm{Hz})$ **(14)** $1.50\,\mathrm{MJ}\div20.0\,\mathrm{m}$

Chapter 3

Logic

Solve the quiz on page 13 again.

Quiz

💪**1.** **(1)** If $mr\omega^2 = 10.0\,\mathrm{kg\cdot m/s^2}$, $m = 5.0\,\mathrm{kg}$, and $r = 1.0\,\mathrm{m}$, what is ω?

(2) If $v_0 t + \frac{1}{2}at^2 = 5.0\,\mathrm{m}$, where $a = -4.0\,\mathrm{m/s^2}$ and $v_0 = 7.0\,\mathrm{m/s}$, what is t?

(3) At time $t = 0$, a particle is at $(x, y) = (2.0\,\mathrm{m}, 0)$. It moves with a constant velocity $\left(v_x, v_y\right) = (3.0, 4.0)\,\mathrm{m/s}$. What is its position at $t = 1.0\,\mathrm{s}$?

The answers are obviously

$\omega = \pm 1.4\,\mathrm{s^{-1}}$ $\quad$ $t = 1.0\,\mathrm{s}, 2.5\,\mathrm{s}$ $\quad$ $x = 5.0\,\mathrm{m}, y = 4.0\,\mathrm{m}$

but **what are the correct meaning of these commas or the symbol "±" ?**

Quiz

💪**2.** Which are the correct meaning? Guess it.

- [ω is both $+1.4\,\mathrm{s^{-1}}$ and $-1.4\,\mathrm{s^{-1}}$] v.s. [ω is either $+1.4\,\mathrm{s^{-1}}$ or $-1.4\,\mathrm{s^{-1}}$].
- [t is both $1.0\,\mathrm{s}$ and $2.5\,\mathrm{s}$] v.s. [t is either $1.0\,\mathrm{s}$ or $2.5\,\mathrm{s}$].
- [x is $5.0\,\mathrm{m}$ and y is $4.0\,\mathrm{m}$] v.s. [Either x is $5.0\,\mathrm{m}$, or y is $4.0\,\mathrm{m}$].

In elementary educations, these differences are often ignored because kids do not know **logical thinking**; now, you need to do it, as you are a grown-up university student.

3.1 Logic

3.1.1 Statements and truth values

A **statement** is a sentence that is either true or false—not both, not neither. We call this its **truth value**: **true** (T) or **false** (F).

For example:

- "$2 + 2 = 4$" is a statement. Its truth value is T.
- "$3 > 5$" is a statement. Its truth value is F.
- "$\sin(\pi) = 0$" is a statement. Its truth value is T.

We use capital letters P, Q, R, … as short names for statements.

Remark: "Please close the door" and "Is it raining?" are not statements, because they are neither true nor false.

3.1.2 Logical connectives: AND, OR, NOT

Given statements P and Q, we form new statements with **logical connectives**.

Conjunction ($P \wedge Q$, read "P AND Q") is true **only when both** P and Q are true.

Disjunction ($P \vee Q$, read "P OR Q") is true **when at least one** of P and Q is true.

Negation ($\neg P$, read "NOT P") is true when P is false, and false when P is true.

The **truth table** below summarises all cases:

P	Q	$P \wedge Q$	$P \vee Q$	$\neg P$
T	T	T	T	F
T	F	F	T	F
F	T	F	T	T
F	F	F	F	T

Be careful: In everyday English, "or" often means **exclusive or** (one or the other, but not both). In mathematics and physics, "or" always means **inclusive or**: $P \vee Q$ is true even when both P and Q are true.

Quiz

★**3.** Let P be "6 is even" (T) and Q be "6 is divisible by 4" (F). Find the truth values of $P \wedge Q$, $P \vee Q$, $\neg P$, and $\neg Q$.

3.1.3 Open sentences

An expression such as "$x^2 = 4$" or "$x > 3$" contains a variable x. Its truth value depends on the value of x—it is not T or F by itself. We call such an expression an **open sentence**.

Once we fix a value of x, an open sentence becomes a statement with a definite truth value:
- "$x^2 = 4$" is T when $x = 2$ or $x = -2$, and F for any other real x.
- "$x > 3$" is T when $x = 5$, and F when $x = 1$.

Open sentences also combine with AND, OR, and NOT—but only after we fix the value of the variable (or specify the set of values we care about).

Remark: In high school, you wrote equations with the silent assumption that each line is true. For instance, solving $x^2 - 5x + 6 = 0$, you wrote

$$x^2 - 5x + 6 = 0 \quad \Rightarrow \quad (x-2)(x-3) = 0 \quad \Rightarrow \quad x = 2 \quad \text{or} \quad x = 3. \tag{3.1}$$

Each line is an open sentence, and "⇒" says: any x that makes the left side true also makes the right side true. You were doing logical reasoning—just without saying so.

Now we can answer the quiz from the previous page.

- $\omega = \pm 1.4\ \mathrm{s}^{-1}$ means $\left(\omega = +1.4\ \mathrm{s}^{-1}\right)$ OR $\left(\omega = -1.4\ \mathrm{s}^{-1}\right)$. Both values satisfy the original equation, but ω has one specific value—either one.
- $t = 1.0\ \mathrm{s},, 2.5\ \mathrm{s}$ means $(t = 1.0\ \mathrm{s})$ OR $(t = 2.5\ \mathrm{s})$.
- $(x, y) = (5.0\ \mathrm{m},, 4.0\ \mathrm{m})$ means $(x = 5.0\ \mathrm{m})$ AND $(y = 4.0\ \mathrm{m})$ simultaneously.

The comma between separate values means OR; the comma inside a coordinate tuple means AND.

Quiz

★**4.** State the truth value of each open sentence for the given value of x.

(1) $x^2 = 4$ when $x = -2$ **(2)** $x^2 = 4$ when $x = 3$

(3) $x > 0$ when $x = -1$ **(4)** $x^2 > x$ when $x = 2$

★**5.** Let $P(n)$ be "n is even" and $Q(n)$ be "n is divisible by 4". Find the truth values of $P(n) \wedge Q(n)$, $P(n) \vee Q(n)$, and $\neg P(n)$ when (i) $n = 6$ and (ii) $n = 8$.

3.1.4 What does "solve" mean?

You have solved many equations. But what does it mean exactly?

To **solve** an equation for x means: find **all** values of x that make the equation true—no more, no less.

"No more, no less" has two parts:

- **No less** (**exhaustiveness**): do not miss any solution.
- **No more** (**no extraneous solutions**): do not include values that do not satisfy the equation.

Both errors are common. Let us look at each.

Example 3.5 (Missing a solution)

Solve $x^2 = 4$.

Solution

The equation $x^2 = 4$ is true when $x = 2$ and also when $x = -2$, and false for every other real x. So the complete answer is: $x = 2$ OR $x = -2$, often written $x = \pm 2$.

Writing only $x = 2$ is **incomplete**: it misses $x = -2$. □

Be careful: Taking a square root does not simply give $x = \sqrt{4} = 2$. It gives $|x| = 2$, which means $x = 2$ OR $x = -2$. Forgetting the negative root is one of the most common errors in high-school and university physics.

The set of all solutions is called the **solution set**. A correct solution is one whose solution set matches exactly.

Example 3.6 (Extraneous solution)

Solve $\sqrt{x+2} = x$.

Solution

Squaring both sides: $x + 2 = x^2$, so $x^2 - x - 2 = 0$, giving $(x-2)(x+1) = 0$, i.e., $x = 2$ or $x = -1$.

But squaring can introduce false solutions, so we must check:

- $x = 2$: $\sqrt{4} = 2$. ✓
- $x = -1$: $\sqrt{1} = 1 \neq -1$. ✗

The solution set is $\{2\}$ only. □

Remark: Why did $x = -1$ appear? Squaring $\sqrt{x+2} = x$ gives the same equation as squaring $\sqrt{x+2} = -x$. So we solved a slightly different (broader) equation by accident, and picked up an extra solution. Checking is not optional—it is part of the solution.

Quiz

★**6.** A student solves $x^2 - 3x = 0$ by dividing both sides by x and gets $x = 3$. What is wrong? Find the complete solution set.

★**7.** Solve $|x - 1| = 3$. Verify both answers.

3.1.5 Assumptions, definitions, and conclusions

A large source of confusion in university physics is mixing up three types of statements:

Assumption (**hypothesis**)	A statement we **declare to be true** for the purpose of an argument. Example: "Let $m = 2.0$ kg." We do not prove it; we simply accept it.
Definition	A statement that **gives meaning** to a symbol or concept. Example: "Let $v := \frac{\mathrm{d}x}{\mathrm{d}t}$." A definition is true by construction; it cannot be wrong.
Conclusion (**deduction**)	A statement that is **derived** from assumptions and definitions by logical steps. Example: "Therefore $v = 3.0$ m/s." Its truth depends on the truth of the assumptions.

Remark: In a calculation, every line is one of these three types. Being clear about which type each line is will prevent many errors.

Here is a typical example of how the three types appear together:

Example 3.7 (Types of statements in a calculation)

A particle moves as $x(t) = 3t^2 + 1$ m. Find the velocity at $t = 2.0$ s.

Solution

- **Definition**: Let $v(t) := \frac{\mathrm{d}x}{\mathrm{d}t}$. (This defines what "velocity" means here.)
- **Deduction**: $v(t) = 6t$ m/s. (Derived by differentiating.)
- **Assumption**: $t = 2.0$ s. (Given in the problem.)
- **Deduction**: $v(2.0\ \mathrm{s}) = 12$ m/s. (Substituting the assumption.) □

Be careful: A common mistake is to treat a **conclusion** as if it were an **assumption**. For example, from $x^2 = 9$ you can deduce $x = 3$ or $x = -3$. But you **cannot** then assume $x = 3$ without justification and use it to derive further results—unless the problem or context confirms it. The solution $x = 3$ is a conclusion, not a new assumption.

Quiz

★**8.** In the following solution, label each line as "assumption", "definition", or "deduction".

> A spring has spring constant $k = 50$ N/m. A block of mass $m = 2.0$ kg is attached. Let $\omega := \sqrt{k/m}$. Then $\omega = 5.0\ \mathrm{s}^{-1}$. The period is $T = 2\pi/\omega = 1.26$ s.

3.1.6 Assumptions change conclusions

The previous section defined assumptions and conclusions in the abstract. Let us see what happens in practice when we change the assumptions.

Consider the equation

$$ax^2 + bx + c = 0. \tag{3.2}$$

A student is asked to "solve for x." What does that mean? It means: **given assumptions about a, b, c, deduce the value(s) of x.**

Example 3.8 (Solving a quadratic---carefully)

Solve Eq. (3.2) for x.

Solution

We must state our assumptions before deducing anything.

Case 1. Assume $a = 0$ and $b = 0$ and $c = 0$. Then every x satisfies Eq. (3.2). The conclusion is: x can be any real number.

Case 2. Assume $a = 0$ and $b = 0$ and $c \neq 0$. Then Eq. (3.2) becomes $c = 0$, which contradicts $c \neq 0$. The conclusion is: there is no solution.

Case 3. Assume $a = 0$ and $b \neq 0$. Then Eq. (3.2) becomes $bx + c = 0$, so $x = -\frac{c}{b}$. The conclusion is: $x = -\frac{c}{b}$.

Case 4. Assume $a \neq 0$. Then we may divide by a. Completing the square gives

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}, \tag{3.3}$$

provided $b^2 - 4ac \geq 0$. (If $b^2 - 4ac < 0$, there is no real solution.) □

Be careful: Many students jump straight to Case 4 and write $x = \frac{-b\pm\sqrt{b^2-4ac}}{2a}$ without checking whether $a = 0$. This is a logical error: the formula is a **conclusion** that is only valid **under the assumption** $a \neq 0$. If $a = 0$, the formula gives division by zero —which is undefined.

In physics, variables like a or m are often assumed nonzero without comment, because the physical context makes it obvious. But in mathematics, and in any careful argument, you must state every assumption explicitly before you draw a conclusion from it.

This is the essence of logical thinking in problem-solving:

Every conclusion is valid only under specific assumptions.
Change the assumptions, and the conclusion may change.

Quiz

★**9.** A student writes the following solution to "solve $mx = F$ for x":

$$x = \frac{F}{m}.$$

What assumption is missing? What happens if that assumption is violated? Write a complete solution that covers all cases.

*****10.** Solve $x^2 = k$ for x, carefully stating all assumptions and covering all cases (consider $k > 0$, $k = 0$, $k < 0$).

3.1.7 Implication

A **conditional statement** (also called an **implication**) has the form:

If P, then Q. (written $P \Rightarrow Q$)

Here P is the **hypothesis** (or condition) and Q is the **conclusion**.

The implication $P \Rightarrow Q$ is false only when P is true and Q is false—you cannot have a true hypothesis lead to a false conclusion.

P	Q	$P \Rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

> **Remark:** When P is false, $P \Rightarrow Q$ is considered true regardless of Q. This may feel strange. Think of the promise: "If it rains, I will bring an umbrella." If it does not rain, you have not broken the promise—no matter what you do.

From $P \Rightarrow Q$ we define three related statements:

Name	Statement
implication	$P \Rightarrow Q$
converse	$Q \Rightarrow P$
contrapositive	$\neg Q \Rightarrow \neg P$
inverse	$\neg P \Rightarrow \neg Q$

The implication and its contrapositive are **logically equivalent**: $P \Rightarrow Q$ is true exactly when $\neg Q \Rightarrow \neg P$ is true. The converse and inverse are also equivalent to each other, but they are **not** equivalent to the original implication.

Example 3.9 (Contrapositive)

Let P be "n is divisible by 4" and Q be "n is even".

- Implication: "If n is divisible by 4, then n is even." (True.)
- Contrapositive: "If n is not even, then n is not divisible by 4." (True—and equivalent to the implication.)
- Converse: "If n is even, then n is divisible by 4." (False: $n = 6$ is a counterexample.)

Solution

We verify only the implication. If $4 \mid n$, then $n = 4k$ for some integer k, so $n = 2(2k)$—hence n is even. □

> **Be careful:** The converse $Q \Rightarrow P$ is **not** generally equivalent to $P \Rightarrow Q$. Assuming the converse is true is a very common mistake in physics reasoning.

Implication connects naturally to equation solving. When you transform an equation step by step, you are writing a chain of implications. There are two kinds of step, and they are not the same:

- A step is $\Rightarrow$ (one-way) if the new equation follows from the old one, but not necessarily vice versa.

- A step is $\Leftrightarrow$ (two-way, **equivalent**) if the two equations are true for exactly the same values of x.

For example:

$$\begin{aligned} x = 3 \quad &\Rightarrow \quad x^2 = 9 && \text{(squaring: one-way)} \\ x^2 = 9 \quad &\Rightarrow \quad x = 3 \quad \text{or} \quad x = -3 && \text{(taking square root)} \end{aligned} \tag{3.4}$$

The first step is only $\Rightarrow$, not $\Leftrightarrow$: squaring loses the sign of x. A chain of $\Leftrightarrow$ steps is ideal because it means your solution set is exactly correct. A chain containing even one $\Rightarrow$ step means you may have introduced **extraneous solutions** and must check each answer.

Example 3.10 (Extraneous solutions)

Solve $\sqrt{x+2} = x$.

Solution

$$\begin{aligned} \sqrt{x+2} = x \quad &\Rightarrow \quad x + 2 = x^2 \quad \text{(squaring: only =>)} \\ &\Leftrightarrow \quad x^2 - x - 2 = 0 \\ &\Leftrightarrow \quad (x-2)(x+1) = 0 \\ &\Leftrightarrow \quad x = 2 \quad \text{or} \quad x = -1. \end{aligned} \tag{3.5}$$

Because we used $\Rightarrow$ in the first step, we must check both candidates.

- $x = 2$: $\sqrt{4} = 2$. ✓
- $x = -1$: $\sqrt{1} = 1 \neq -1$. ✗ (extraneous)

The only solution is $x = 2$. □

Be careful: When a step is only $\Rightarrow$, you do not lose solutions—you may **gain** false ones. When a step is only $\Leftarrow$ (the reverse direction fails), you may **lose** solutions. Multiplying both sides by an expression that could be zero is a common source of $\Leftarrow$ -only steps.

Quiz

★11. In each transformation below, state whether the step is $\Leftrightarrow$, $\Rightarrow$ only, or $\Leftarrow$ only (for real x). Justify briefly.

(1) $x - 1 = 0 \quad ? \quad x = 1$.

(2) $x^2 = 1 \quad ? \quad x = 1$.

(3) $x(x-1) = 0 \quad ? \quad x - 1 = 0$.

(4) $(x-2)^2 = 0 \quad ? \quad x = 2$.

3.1.8 Necessary and sufficient conditions

When $P \Rightarrow Q$, we say:

- P is a **sufficient condition** for Q: knowing P is enough to conclude Q.
- Q is a **necessary condition** for P: if Q is false, P cannot be true.

When $P \Rightarrow Q$ **and** $Q \Rightarrow P$ both hold, we write $P \Leftrightarrow Q$ ("P if and only if Q", abbreviated **iff**). In this case, P and Q are **logically equivalent**, and P is both necessary and sufficient for Q.

Example 3.11 (Necessary vs sufficient)

Let n be an integer.

- "n is divisible by 4" $\Rightarrow$ "n is even": divisibility by 4 is **sufficient** for n to be even.
- "n is even" is **necessary** for "n is divisible by 4": an odd number can never be divisible by 4.
- But the converse fails (as seen above), so they are not equivalent.

Solution

Already discussed. The key point: sufficient $\neq$ necessary. □

Quiz

★**12.** For each pair, decide whether the first condition is sufficient, necessary, both (iff), or neither for the second.

(1) "$x = 3$" and "$x^2 = 9$".

(2) "$x > 0$ and $y > 0$" and "$xy > 0$".

3.1.9 Quantifiers

Many mathematical statements involve **quantifiers**, which tell us **how many** objects satisfy a condition.

The **universal quantifier** $\forall$ means "for all" (or "for every"):

$$\forall x, P(x) \quad \text{means} \quad \text{"}P(x) \text{ is true for every } x\text{".}$$

The **existential quantifier** $\exists$ means "there exists" (or "for some"):

$$\exists x, P(x) \quad \text{means} \quad \text{"there is at least one } x \text{ for which } P(x) \text{ is true".}$$

The **negation** of quantified statements follows De Morgan's rules:

$$\begin{aligned} \neg(\forall x, P(x)) &\Leftrightarrow \exists x, \neg P(x), \\ \neg(\exists x, P(x)) &\Leftrightarrow \forall x, \neg P(x). \end{aligned} \tag{3.6}$$

In words: to disprove a "for all" statement, you only need **one counterexample**. To disprove a "there exists" statement, you must show the property fails for **every** object.

Example 3.12 (Negating a quantified statement)

Consider the statement "Every real number has a positive square root."

Formally: $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}, y > 0 \wedge y^2 = x$.

This is false. Its negation is: $\exists x \in \mathbb{R}, \forall y \in \mathbb{R}, y \leq 0 \vee y^2 \neq x$. One counterexample: $x = -1$ has no real square root.

Solution

We only need to exhibit one x for which no such y exists. Take $x = -1$. For any real y, $y^2 \geq 0 > -1$, so $y^2 \neq x$. □

Be careful: The order of quantifiers matters. $\forall x, \exists y, P(x, y)$ is **not** the same as $\exists y, \forall x, P(x, y)$. The first says "for each x, we can find a (possibly different) y"; the second says "there is one fixed y that works for all x".

Problems

★**3.1.** Let $P(x)$ be "$x^2 > 0$". Write the negation of each statement and decide whether the original and the negation are true (for $x \in \mathbb{R}$).

(1) $\forall x, P(x)$ **(2)** $\exists x, \neg P(x)$

*****3.2.** Show that "If $a^2 = b^2$ then $a = b$" is false by giving a counterexample. Write the contrapositive, the converse, and the inverse. Determine which of the four statements are true.

*****3.3.** Suppose $P \Rightarrow Q$ and $Q \Rightarrow R$. Prove $P \Rightarrow R$. (This is called the **law of syllogism**.)

****3.4.** Let P, Q, R be propositions. Using truth tables or logical rules, show that

$$(P \Rightarrow Q) \wedge (P \Rightarrow R) \quad \Leftrightarrow \quad P \Rightarrow (Q \wedge R). \tag{3.7}$$